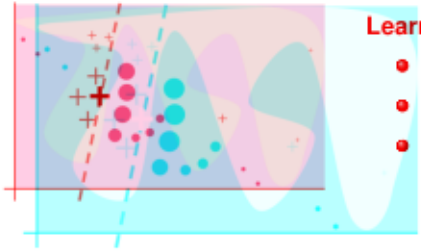
# Interpretable Machine Learning

## Introduction to Local Explanations



**Learning goals**

- Understand motivation for local explanations
- Develop an intuition for possible use-cases
- Know characteristics of local explanation methods

# METHODOLOGICAL MOTIVATION

- Purpose of local explanations:
  - Insight into the driving factors for a **particular decision**
  - Understand the ML model's decisions in a **local neighborhood** of a given input
    (e.g., feature vector)



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- Local Methods can address questions such as:
  - **Why** did the model decide to predict $\hat{y}$ for input $\mathbf{x}$?
  - **How** does the model decide for observations that are similar to $\mathbf{x}$?
  - **What** would the ML model have decided if $\mathbf{x}$ its values in $\mathcal{X}$ were different?
  - **Where** (in which regions in $\mathcal{X}$) does the model fail?

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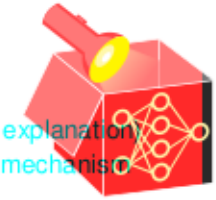
# SOCIAL MOTIVATION



- Explanations for laypersons must be tailored to the **explainee** (who receives the explanation)
  $\rightsquigarrow$ **case specific**, **easy** for humans to understand, and **faithful** to the explained mechanism



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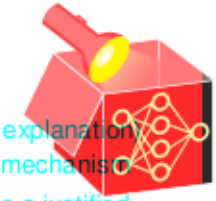

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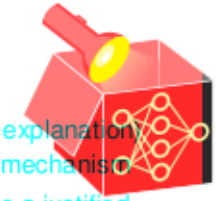

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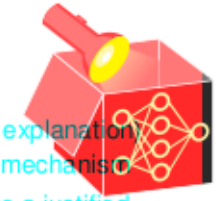

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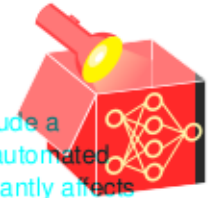
# GDPR: THE RIGHT TO EXPLANATION



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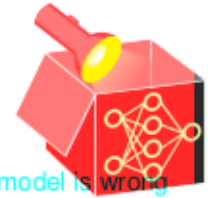
# EXAMPLE: HUSKY OR WOLF?

- We trained a model to predict if an image shows a wolf or a husky
- Below the predictions on six test images are given
- Do you trust our predictor?

**Source:** [Sameer Singh 2018]



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- Can you guess the pattern the ML model learned to identify a wolf?


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# EXAMPLE: HUSKY OR WOLF? USING LIME

- Local explanations highlight the parts of an image which led to the prediction

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# EXAMPLE: LOAN APPLICATION

**Source:** [https://www.elte.hu]

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- Imagine: You apply for a loan at an online bank and are immediately rejected without reasons.
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  - You work at a car company that develops image classifiers for autonomous driving
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# EXAMPLE: STOP OR RIGHT-OF-WAY?

- Imagine:
  - You work at a car company that develops image classifiers for autonomous driving
  - You show your model the following image (an adversarial example)
  - Classifier is 99% sure it describes a right-of-way sign
- Would you entrust other peoples lives into the hands of this software?

**Source:** [Eykholt et. al 2018]

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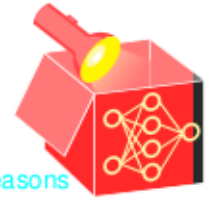

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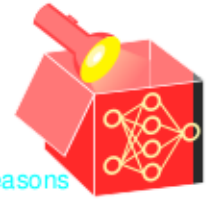

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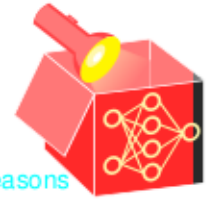

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- **Special:** Due to audience, strong interactions with social sciences and strong connections to cognitive science and neurosciences due to data types


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# CREDIT DATASET

- We illustrate local explanation methods on the German credit data
- 522 complete observations, 9 features containing credit and customer information
- Binary target "risk" indicates if a customer has a 'good' or 'bad' credit risk
- We merged categories with few observations

| name | type | range |
|---|---|---|
| age | numeric | [19, 75] |
| sex | factor | {male, female} |
| job | factor | {0, 1, 2, 3} |
| housing | factor | {free, own, rent} |
| saving.accounts | factor | {little, moderate, rich} |
| checking.accounts | factor | {little, moderate, rich} |
| credit.amount | numeric | [276, 18424] |
| duration | numeric | [6, 72] |
| purpose | numeric | {others, car, furniture, radio/TV} |
| risk | factor | {good, bad} |